

EE 308 Lab 3

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Performance

Indented Frequency Band	Measured Lower Bound	Measured Upper Bound	Measured Frequency Range	Band Overlap
2Hz – 20Hz	1.905Hz	24.257Hz	22.352Hz	8.307Hz
20Hz – 200Hz	15.95Hz	211.02Hz	195.07Hz	51.69Hz
200Hz – 2kHz	159.330Hz	2.128kHz	1.969kHz	472Hz
2kHz – 20kHz	1.656kHz	21.414kHz	19.758kHz	5.346kHz
20kHz – 200kHz	16.068kHz	204.52kHz	188.452kHz	17.51kHz
200kHz – 2MHz	187.01kHz	2.11MHz	1.923MHz	NA

Indented Frequency Band	Duty Cycle Error	Feedthrough Error	Amplitude Variation
2Hz – 20Hz	-0.33%	No measurable feedthrough error	1.11V to 1.11V
20Hz – 200Hz	-0.34%	No measurable feedthrough error	1.11V to 1.11V
200Hz – 2kHz	-0.34%	No measurable feedthrough error	1.11V to 1.13V
2kHz – 20kHz	-0.30%	No measurable feedthrough error	1.15V to 1.13V
20kHz – 200kHz	-0.07%	No measurable feedthrough error	1.13V to 1.15V
200kHz – 2MHz	+2.3%	42.5mV to 595mV	1.15V to 1.25V

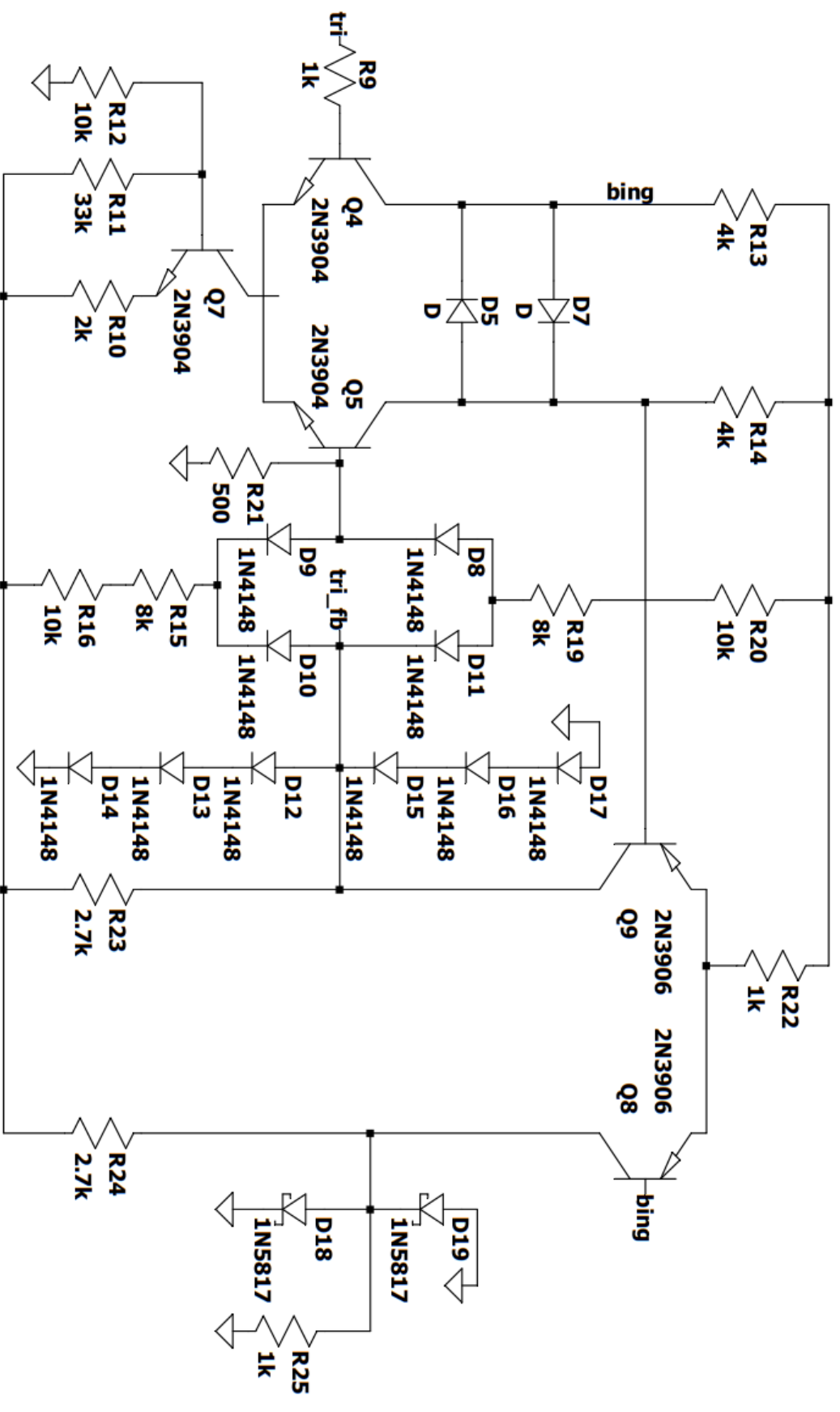
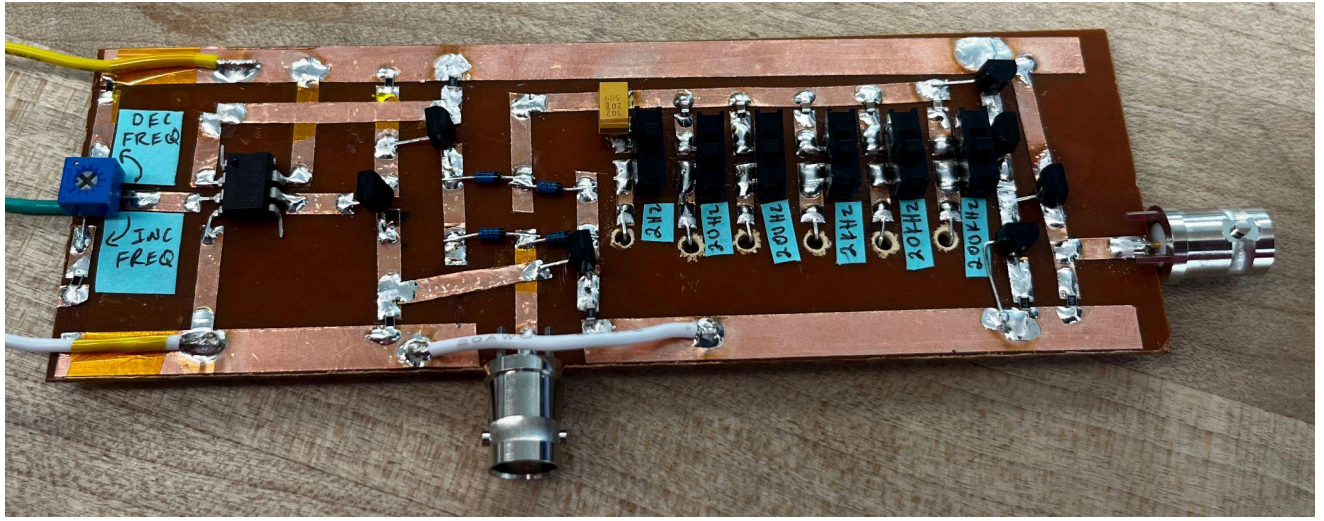


Photo of board



Technical Discussion

In designing the integrator we first chose our capacitor ranges for the charging and discharging of the triangle wave. From commercial examples ranges typically start around 1uF to 10uF and end around 100pF to 1nF. We were worried about the parasitic capacitances from the diodes and copper tape making the timing capacitors too high for the range so we started our capacitor range from ~10uF so that the timing capacitor values would not be affected by any parasitics in the area. Our final calculated capacitor range and their intended frequency range was the following.

Frequency Range	Timing Capacitor Value
2Hz - 20Hz	10uF
20Hz - 200Hz	1uF
200Hz - 2kHz	100nF
2kHz - 20kHz	10nF
20kHz - 200kHz	1nF
200kHz - 2MHz	100pF

The feedback voltage from the Schmitt trigger was roughly 2Vpp so from this we can calculate the current required to ramp up and down. For each frequency range the current min and max should be the same with the minimum current required at the lower frequency range and the maximum current required at the upper frequency range. Because of this we only need to calculate the currents for one range and use these results for all the frequency range.

Let's use the lowest range as a starting point. We know the capacitor value will be 10uF and the peak voltage is 1Vpp. The ramp time is half of the period of the frequency of interest so using these values we can calculate the required voltage. The minimum current is then the following.

$$T_{2Hz} = 0.25s \times 2$$
$$i_c = C \frac{dV}{dt} \approx C \frac{\Delta V}{\Delta t} = 10uF \frac{1}{0.25s} = 0.04mA$$

The maximum current is then the following

$$T_{20Hz} = 0.025s \times 2$$
$$i_c \approx C \frac{\Delta V}{\Delta t} = 10uF \frac{1}{0.025s} = 0.4mA$$

Now we can consider our frequency adjustment method. We use a bounded 10k potentiometer to adjust our frequency. From the structure of our schematic, the output of the potentiometer is fed to the noninverting terminal of an opamp. The inverting terminal is then connected to a resistor connected to the +12V rail. This is what sets our variable current source. Since the

output voltage of the potentiometer swings from ~1V to ~11V, we use 11V for the lower frequency bound (since it requires less current), and 1V for the upper frequency bound (since it requires more current). Now we can calculate the resistor's value. For the lower frequency range we get a resistor value of the following.

$$R = \frac{12V-11V}{0.04mA} = 25k\Omega$$

Then if we recalculate with the upper frequency bound we get the following.

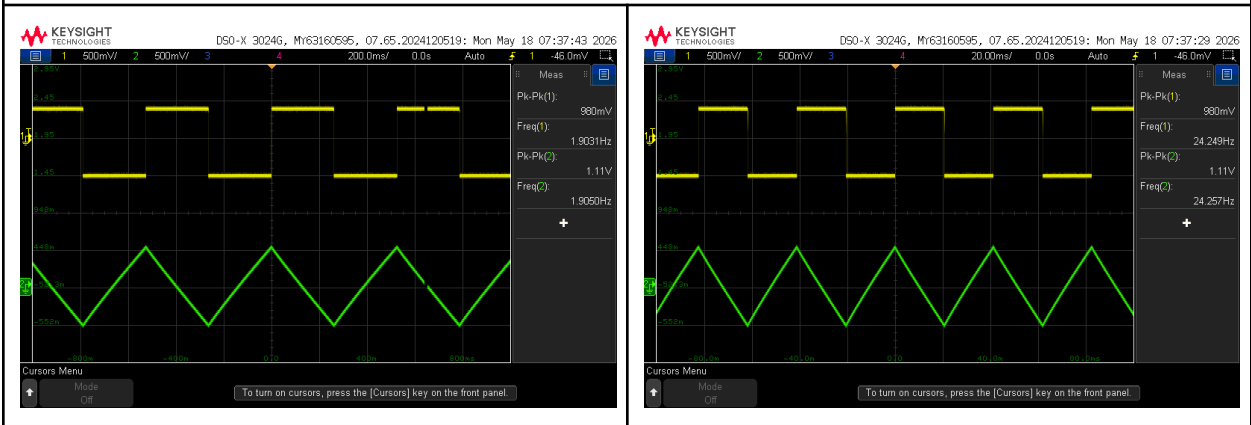
$$R = \frac{12V-1V}{0.4mA} = 27.5k\Omega$$

We ended up using two 25k resistors for the top and bottom current source for simplicity and symmetry. In tuning we had to reduce the capacitor values to meet the necessary specifications. This is likely due to added parasitics from the copper tape and diodes.

Between the output of the integrator and the Schmitt trigger, we built a simple buffer so that the timing capacitor sees high impedance. The important design specifications for the buffer are to make sure it has a bandwidth high enough to include several times the fundamental frequency to ensure the triangle wave makes it through with minimum information lost. We used a JFET to implement the high impedance, a diode at the source of the JFET to remove a small offset, and a current source made from another JFET and 220 Ω resistor to set the current to 8mA. Finally, there is a follower stage with the current set to 26mA via an emitter resistor.

Triangle wave at high and low end of each frequency band

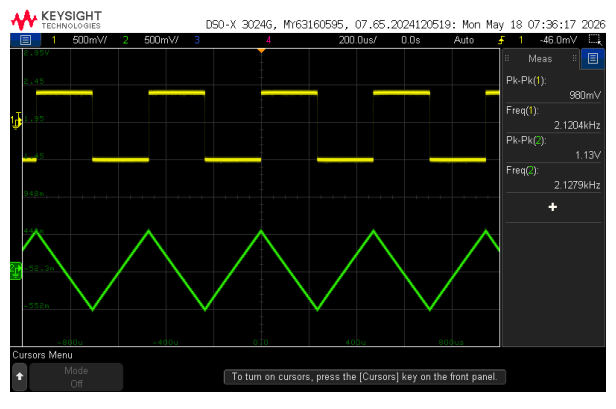
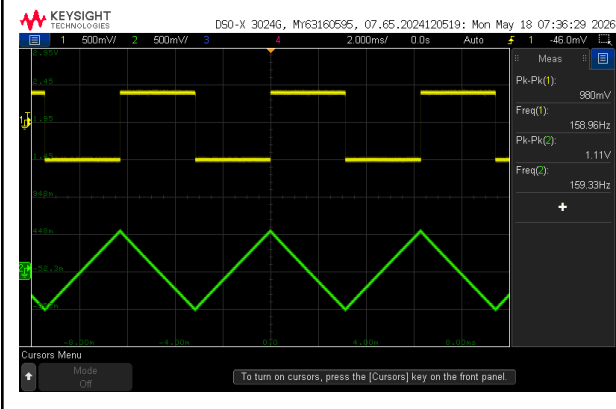
Frequency Band: 2Hz to 20Hz



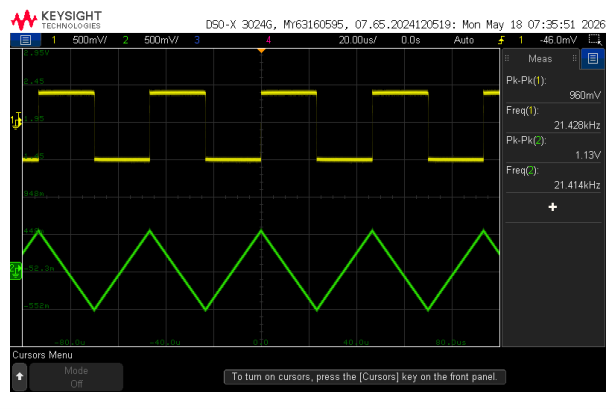
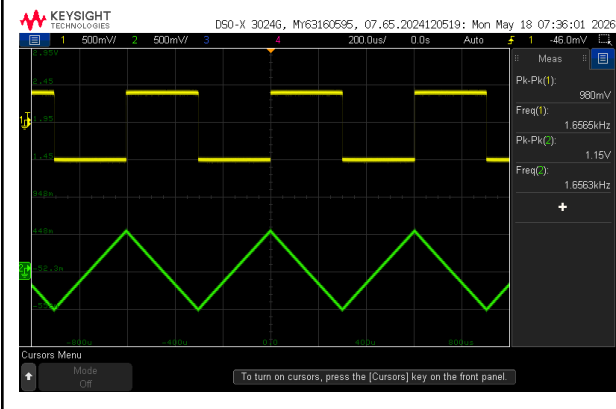
Frequency Band: 20Hz to 200Hz



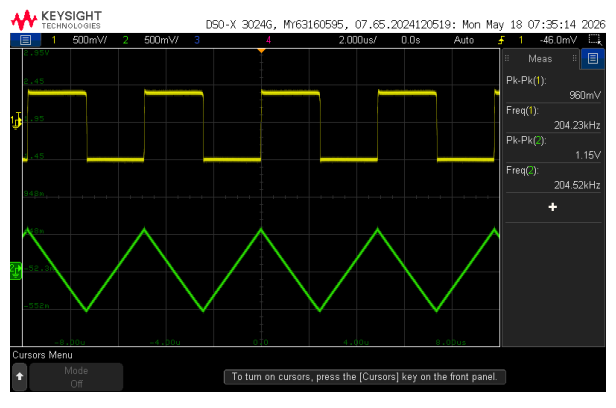
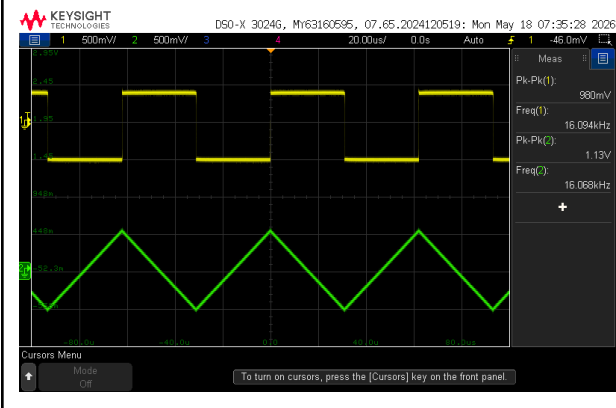
Frequency Band: 200Hz to 2kHz



Frequency Band: 2kHz to 20kHz



Frequency Band: 20kHz to 200kHz



Frequency Band: 200kHz to 2MHz

